

Fourier Series and Fourier Transform

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1 INTRODUCTION

The primary objective of this laboratory exercise is to mathematically analyze and computationally simulate continuous-time signals using Fourier Series and Fourier Transforms. Analyzing signals in the frequency domain shows the individual sinusoidal frequency components that make up complex waveforms [1].

The exponential Fourier Series specifically decomposes continuous-time periodic signals into an infinite sum of complex exponential functions [2]. Reconstructing a signal using only a finite number of exponential terms results in waveform that exhibits oscillations near discontinuities which is a behavior known as the Gibbs phenomenon [3].

Meanwhile for aperiodic signals the Fourier Transform extends to analyzing the signal over a continuous spectrum of frequencies instead of the discrete harmonics [1]. Applying Fourier Transform to a system's impulse response gives its transfer function. The transfer function dictates how the system filters different input frequencies [2]. Moreover, the operational boundary of this filter is defined by 3dB cut-off frequency that marks the exact point where the signal's power is reduced by half [4].

Several specific MATLAB functions are required to computationally implement and visualize these mathematical operations. The syms and assume functions establish the foundational symbolic variables and their specific mathematical constraints for calculus operations. The int() function calculates the exact analytical integral to find the Fourier coefficients, while sum() and subs() are combined to evaluate and add the truncated series terms. Finally, the fourier() command computes the continuous transfer function, and plotting tools like stem(), plot(), and fplot() construct the visual representations of the signal magnitudes and cut-off relationships.

2 EXPERIMENTAL DESIGN

This section outlines the procedural framework used to decompose periodic signals and analyze system impulse responses using MATLAB's computational tools. The design is structured to

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ensure that the mathematical transition from the time domain to the frequency domain is accurate and replicable.

2.1 Signal Modeling and Fourier Series Decomposition

The first phase of the experiment focuses on representing a periodic sawtooth signal, $x(t)$, through an Exponential Fourier Series. Based on the visual characteristics of the signal provided in the laboratory specifications, the period is identified as $T = 0.75$, which establishes the fundamental frequency as shown in Equation 1:

$$\omega_0 = \frac{2\pi}{T} \quad (1)$$

To reconstruct the signal, the exponential Fourier coefficients (C_n) must be calculated. The methodology employs the following integral expression evaluated over one period:

$$C_n = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jn\omega_0 t} dt \quad (2)$$

The experimental setup utilizes MATLAB's Symbolic Math Toolbox to define the signal $x(t) = -2t$ and solve the integral for a harmonic range of n from -10 to 10 . A critical component of this design involves handling numerical singularities; an infinitesimal constant (eps) is added to the frequency index when $n = 0$ to prevent division-by-zero errors during numerical evaluation.

2.2 System Response and Frequency Analysis

The second phase involves characterizing a system defined by its impulse response. The system is modeled by the function $h(t) = \frac{a}{2} e^{-a|t|}$ where $a \geq 0$. The experimental design uses the fourier function in MATLAB to determine the transfer function, $H(\omega)$, in the frequency domain.

A key performance metric in this design is the 3dB cut-off frequency (ω_c), defined as the point where the magnitude of the transfer function drops to $1/\sqrt{2}$ of its maximum value. The methodology requires solving the following magnitude equation to obtain a symbolic expression for ω_c :

$$|H(\omega)| = \frac{1}{\sqrt{2}} \quad (3)$$

2.3 Step-by-Step Procedure

To ensure the replicability of the results, the following procedural steps are implemented:

- 1) **Symbolic Initialization:** Define t , n , ω , and a as symbolic variables within the MATLAB workspace.
- 2) **Signal Construction:** Define the periodic signal $x(t)$ and extend it to three full periods using `piecewise` or `heaviside` functions to observe signal behavior across the interval $-1 \leq t \leq 1$.
- 3) **Coefficient Calculation:** Utilize the `int` function to compute the symbolic expression for the Fourier coefficients C_n .
- 4) **Numerical Reconstruction:** Convert symbolic coefficients to numerical values and reconstruct the signal $x_{pN}(t)$ by summing the harmonics from $n = -10$ to 10 .
- 5) **Signal Comparison:** Plot the original signal against the truncated Fourier series using a time increment of $\Delta t = 1/1000$ to visualize the approximation and the effects of the Gibbs phenomenon.
- 6) **System Characterization:** Define the impulse response $h(t)$, compute the Fourier Transform $H(\omega)$, and solve for the cut-off frequency ω_c .
- 7) **Parametric Sweep:** Generate magnitude response plots for $|H(\omega)|$ across varying system parameters $a = [10, 50, 80, 100]$ to observe bandwidth variations.

2.4 Computational Logic

The programmatic flow of the experiment is summarized in the following logic:

- Initialize symbolic environment and define fundamental signal parameters (T , ω_0).
- Evaluate the integral for C_n and store coefficients in a vector array.
- Loop through the harmonic indices to compute the partial sum of the Fourier series.
- Apply the Fourier Transform to the system impulse response and solve for ω_c using the `solve` command.
- Visualize the magnitude of C_n , the reconstructed time-domain signal, and the frequency-domain magnitude responses.

3 ANALYSIS OF RESULTS

The results obtained from this exercise provide a comprehensive understanding of frequency-domain representations and system filtering behavior. To ensure computational reliability and mathematical validity of the exercise, independent simulations are consolidated and verified.

3.1 Fourier Series Decomposition and Magnitude Spectrum

A periodic signal can be broken down into individual frequency components having the magnitude spectrum showing how the signal's energy is distributed across the mentioned frequencies. The frequency-domain representation of the given periodic sawtooth signal $x_p(t) = -2t$ with a period of $T = 0.75$ quantifies energy distribution. Through calculating the exponential Fourier series coefficients (C_n) the analytical composition of the signal is found. The resulting magnitude spectrum which was plotted for the harmonic index $-10 \leq n \leq 10$ reveals a symmetric decay centered around the fundamental DC component. This peak shows that the lowest frequencies hold the majority of the

waveform's power while the higher harmonics have less. Just to maintain computational stability throughout numerical evaluation of this an infinitesimal variable (`eps`) was required to prevent division-by-zero singularities at the exact zero-frequency index ($n = 0$).

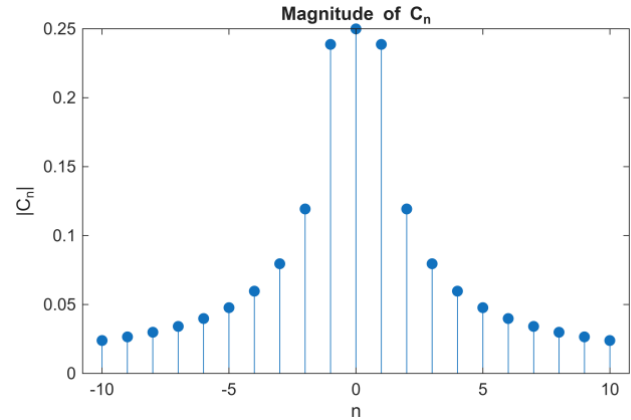


Figure 1. Magnitude spectrum of the exponential Fourier series coefficients for $-10 \leq n \leq 10$, illustrating peak energy concentration at the fundamental frequency.

3.2 Signal Reconstruction and the Gibbs Phenomenon

Reconstructing a signal that contains amplitude changes using a limited number of frequencies results in a behavior known as the Gibbs phenomenon. Generating the analytical expression for the truncated signal $x_{pN}(t)$ by utilizing the computed finite series of 21 harmonics produces algebraic complexity and the time-domain also needs numerical substitution over time vector $\Delta t = 0.001$. Overlaying this reconstructed sequence against the ideal original signal $x_p(t)$ shows the mathematical limitations of finite harmonic summation while the approximation accurately follows the linear downward slope of the sawtooth wave. This exhibits significant structural deviations at the interval boundaries.

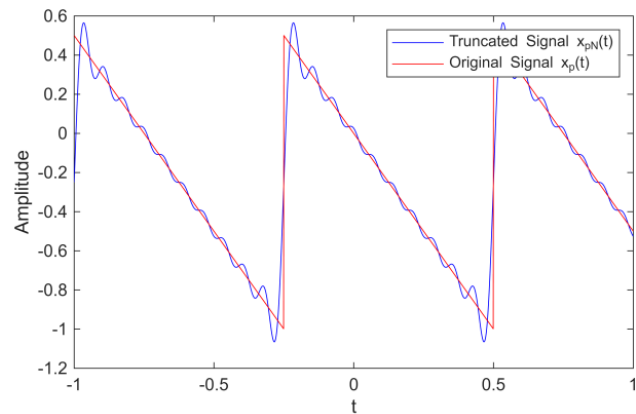


Figure 2. Time-domain comparison of the original sawtooth signal and its truncated Fourier reconstruction, highlighting oscillatory ringing at waveform discontinuities.

3.3 Transfer Function and Cut-off Linearity

A system's transfer function dictates its frequency filtering behavior and the 3dB cut-off frequency in the exercise defines the exact boundary where the transmitted signal loses half of its

power. For aperiodic continuous-time systems the exponential impulse response $h(t) = \frac{a}{2}e^{-a|t|}$ represents an ideal low-pass filter. Deriving the transfer function $H(\omega)$ using continuous Fourier transform establishes the baseline frequency-domain behavior. Analytically solving for the 3dB cut-off frequency yields multiple algebraic roots due to the quadratic nature of the frequency variable and getting the magnitude establishes the exact mathematical relationship $\omega_c = a\sqrt{\sqrt{2}-1}$. Evaluating and plotting that magnitude expression through $0 \leq a \leq 100$ generates a strictly positive linear plot. This graphical result mathematically proves a direct proportional dependency with a scaling constant of approximately 0.6436 where the operational frequency boundary of the filter is defined by its exponential scaling factor a .

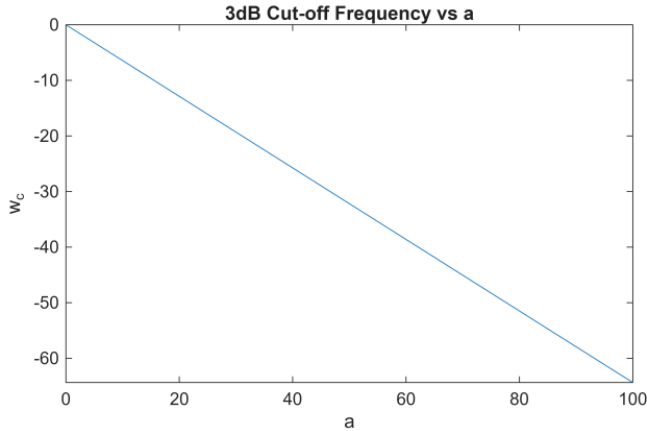


Figure 3. Linear proportional relationship between the exponential system parameter a and the resulting positive 3dB cut-off frequency magnitude ω_c .

3.4 Parametric Bandwidth Expansion

Increasing a low-pass filter's cut-off frequency naturally widens its passband that allowing a broader range of frequencies to pass through the system unattenuated. The practical implications of the previously established linear relationship are validated by evaluating the magnitude response curves $|H(\omega)|$ at discrete parametric intervals of $a = 10, 50, 80$, and 100 . As the defining parameter a scales upward the attenuation slopes relaxes causing the bell-shaped passband to widen uniformly outward across the frequency axis. Consequently, higher cut-off frequencies natively expand the system's absolute bandwidth permitting a broader continuous spectrum of input frequencies to propagate through the system unattenuated [3].

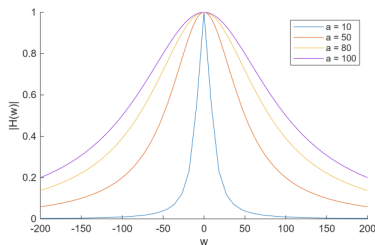


Figure 4. Superimposed magnitude responses demonstrating continuous bandwidth expansion as the exponential parameter a increases from 10 to 100.

4 DISCUSSION AND CONCLUSION

The experimental results validate the application of Exponential Fourier Series (EFS) in decomposing periodic signals and characterizing continuous-time systems. In the first phase, the reconstruction of the sawtooth signal $x(t) = -2t$ using calculated Fourier coefficients C_n demonstrated that as the number of terms n increases, the approximation converges toward the original function. However, the presence of oscillations near the points of discontinuity confirms the existence of the Gibbs phenomenon. As noted in [1], this phenomenon is an inherent mathematical limitation where a finite sum of continuous sine and cosine waves cannot perfectly represent a jump discontinuity, resulting in a persistent overshoot of approximately 9%. This necessitates the consideration of the Dirichlet conditions when determining the representability of a signal in the frequency domain [3].

The second phase of the laboratory focused on the system impulse response $h(t) = \frac{a}{2}e^{-a|t|}$. The data revealed a linear proportional relationship between the exponential scaling factor a and the 3dB cut-off frequency ω_c . As the parameter a increased from 10 to 100, the system's magnitude response $|H(\omega)|$ exhibited a widening passband. This behavior identifies the system as a low-pass filter where a serves as the primary control for bandwidth expansion. Such parametric control is vital in practical engineering applications, as it allows for the precise adjustment of a filter's spectral envelope to either isolate signals of interest or suppress high-frequency noise [2].

In conclusion, the experiment successfully demonstrated the transition from time-domain signal modeling to frequency-domain system analysis. The use of MATLAB's Symbolic Math Toolbox allowed for the accurate calculation of C_n and the visualization of parametric shifts in filter performance. The findings underscore the trade-offs between series truncation and signal fidelity, as well as the direct impact of system parameters on absolute bandwidth. These principles are fundamental to the design of communication systems and signal processing chains where spectral efficiency and reconstruction accuracy are paramount [4].

5 AUTHORS CONTRIBUTION

On what contributions to specify, see the terms at www.elsevier.com.

- 1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration
- 2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision
- 3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing

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